own. Take a phone call, for example: its metadata might include the date and time of the call, the call's duration, the number from which the call was made, the number being called, and their locations. An email's metadata might include information about what type of computer it was generated on, where, and when, who the computer belonged to, who sent the email, who received it, where and when it was sent and received, and who if anyone besides the sender and recipient accessed it, and where and when. Metadata can tell your surveillant the address you slept at last night and what time you got up this morning. It reveals every place you visited during your day and how long you spent there. It shows who you were in touch with and who was in touch with you.

It's this fact that obliterates any government claim that metadata is somehow not a direct window into the substance of a communication. With the dizzying volume of digital communications in the world, there is simply no way that every phone call could be listened to or email could be read. Even if it were feasible, however, it still wouldn't be useful, and anyway, metadata makes this unnecessary by winnowing the field. This is why it's best to regard metadata not as some benign abstraction, but as the very essence of content: it is precisely the first line of information that the party surveilling you requires.

There's another thing, too: content is usually defined as something that you knowingly produce. You know what you're saying during a phone call, or what you're writing in an email. But you have hardly any control over the metadata you produce, because it is generated automatically. Just as it's collected, stored, and analyzed by machine, it's made by machine, too, without your participation or even consent. Your devices are constantly communicating for you whether you want them to or not. And, unlike the humans you communicate with of your own volition, your devices don't withhold private information or use code words in an attempt to be discreet. They merely ping the nearest cell phone towers with signals that never lie.

One major irony here is that law, which always lags behind technological innovation by at least a generation, gives substantially more protections to a communication's content than to its metadata—and yet intelligence agencies are far more interested in the metadata—the activity records that allow them both the "big picture" ability to analyze data at scale, and the "little picture" ability to make perfect maps, chronologies, and associative synopses of an individual person's life, from which they presume to extrapolate predictions of behavior. In sum, metadata can tell your surveillant virtually everything they'd ever want or need to know about you, except what's actually going on inside your head.